

Brain Spy Glioblastoma: A Benchmark of snRNA-seq vs. Spatial Transcriptomics CCC algorithms with Graph-Based Analysis

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Abstract: Glioblastoma (GBM) is a highly aggressive brain tumor with profound intratumoral heterogeneity and strong microenvironmental influence. This heterogeneity complicates both tumor boundary delineation and the mechanistic interpretation of tumor-microenvironment interactions. In this study, we focus on a single, clinically relevant question: can we robustly separate tumor from normal tissue-associated states and characterize tumor-associated signaling using single-nucleus RNA sequencing (snRNA-seq) and spatial transcriptomics from the same GBM cohort? We use the publicly available dataset GSE174554, which profiles human IDH-wild-type GBM specimens under standard-of-care therapy with matched primary and recurrent tumors and includes snRNA-seq and spatial assays.

Our approach has two tightly connected components. First, we frame tumor-normal separation as a node classification problem and train graph neural networks (GNNs) on both snRNA-seq and spatial transcriptomics representations. Cells or spots are represented as nodes with learned expression features, edges are constructed from k-nearest-neighbor graphs in a reduced feature space, and the GNN predicts tumor versus normal. Second, we infer and benchmark cell-cell communication (CCC) signals under tumor-normal grouping using multiple CCC strategies. Due to Python package and dependency constraints when attempting to run multiple CCC tools in a single pip environment, we re-implemented and executed the CCC benchmarking pipeline in R to ensure reproducibility and to obtain results across methods.

Across modalities, GNN-based tumor classification achieved strong performance, with the provided final metrics reaching an Accuracy of 0.880, ROC-AUC of 0.949 on snRNA-seq, and an Accuracy of 0.947, ROC-AUC of 0.871 on spatial transcriptomics. CCC benchmarking revealed that simple ligand-receptor scoring approaches produced dense interaction sets while CellChat produced much sparser tumor-normal networks in this setting, with low overlap between methods. We present tumor-associated ligand-receptor pairs dominated by EGFR- and ECM-related signaling (for example, HSP90AA1->EGFR, VEGFA->EGFR, VCAN->EGFR, TNC->EGFR) as consistently high-scoring candidates. Together, these results support GNNs as an effective tumor-normal discriminator and highlight how CCC conclusions can be strongly method-dependent under tumor-standard grouping.

Introduction

Glioblastoma (GBM) is the most aggressive primary malignant brain tumor in adults and remains extremely difficult to treat. Even with maximal surgical resection followed by radiation and temozolomide, outcomes are poor, and recurrence is common. [1] A key reason for this failure is not only rapid tumor growth, but extensive heterogeneity within the tumor and its surrounding microenvironment. Tumor cells shift between transcriptional states, interact with immune and stromal populations, and adapt under therapeutic pressure. [2] These processes occur locally within

tissue. As a result, understanding GBM requires more than identifying which cell states exist. It also requires understanding how cells interact within their local neighborhoods.

Cell-to-cell communication (CCC) is one mechanism through which these local dynamics emerge. CCC methods aim to infer signaling relationships between cellular groups through ligand receptor interactions derived from gene expression data. In GBM, these interactions are closely tied to immune suppression, angiogenesis, extracellular matrix remodeling, and oncogenic signaling. CCC, therefore, provides a useful link between observed transcriptional states and functional behavior. Because many ligand receptor pairs correspond to known signaling pathways, CCC analyses can also help prioritize interactions with potential therapeutic relevance.

Despite this, CCC inference is highly sensitive to both the underlying data and the method chosen. Single-nucleus RNA sequencing captures high-resolution transcriptional profiles but discards spatial context, even though many signaling interactions depend on physical proximity. Spatial transcriptomics preserves tissue structure and adjacency but is typically sparser and prone to mixed-cell signals at the spot level. In addition, CCC methods differ in their filtering strategies, scoring schemes, and statistical assumptions. As a result, different algorithms can produce substantially different communication networks from the same data. This variability complicates interpretation and raises concerns about the stability of biological conclusions drawn from CCC results.

In this study, we analyze the multi-assay glioblastoma dataset GSE174554 [3] within a tumor-versus-normal framework. We focus on two questions. First, how consistent are CCC networks across methods and modalities when applied to the same biological comparison? Second, can neighborhood-aware models leverage graph structure to distinguish tumors from normal states accurately and produce interpretable spatial predictions?

We hypothesize that CCC networks will vary strongly across algorithms, particularly under tumor versus normal grouping, where interaction counts are limited, and filtering choices have an outsized effect. We further expect spatial transcriptomics to yield smaller and more selective interaction networks than snRNA sequencing, due to increased sparsity and spatial constraints. In parallel, we hypothesize that graph neural networks will robustly classify tumor versus normal states in both modalities, and that spatial models will produce locally coherent predictions when mapped back onto tissue coordinates.

By combining CCC benchmarking with graph-based modeling, this work evaluates both the stability of inferred communication networks and the value of neighborhood information for tumor state prediction in glioblastoma.

Related Work

Cell–cell communication (CCC) inference from single-cell RNA-seq is widely used to translate gene expression data into hypotheses about paracrine signaling, immune–tumor interactions, and microenvironment remodeling. Commonly used tools include CellPhoneDB, which combines curated ligand–receptor knowledge with statistical testing to identify enriched interactions between cell groups, and CellChat, which estimates signaling probabilities using a mass-action-inspired framework and supports pathway-level network analysis [4,5,6,7]. These methods are popular in tumor studies because they provide an interpretable link between transcriptional states and system-level interaction hypotheses.

Importantly, CCC inference is not a single well-defined procedure. Different methods rely on different ligand–receptor databases, filtering strategies, and scoring or statistical models, which can lead to substantially different interaction rankings on the same dataset. A systematic comparison of CCC tools reported high variability in predicted interactions and pathway coverage across methods, underscoring the need for method-aware interpretation. Because our study explicitly compares CCC outputs across multiple approaches, including baseline-style scoring, a SingleCellSignalR-style proxy, and CellChat [9,10,11], we treat benchmarking as an integral part of the analysis rather than assuming CCC results are uniquely defined.

In glioblastoma (GBM), CCC is particularly relevant because tumor evolution is shaped by both tumor-intrinsic programs and tumor-extrinsic influences such as immune infiltration and microenvironment remodeling. The dataset analyzed here, GSE174554 [3], profiles IDH–wild-type GBM under standard-of-care therapy and emphasizes changes in cellular composition and tumor–immune interactions across disease states [8]. This context makes CCC inference useful for prioritizing tumor-microenvironment signaling hypotheses, while also making benchmarking essential due to the number of interacting populations and the sensitivity of CCC predictions to method choice and thresholds.

Graph neural networks (GNNs) are well-suited for spatial and single-cell omics because they integrate molecular features with neighborhood structure, which is inherent to tissue data. In spatial transcriptomics, methods such as SpaGCN and STAGATE construct graphs over nearby spots and learn representations that preserve tissue organization, demonstrating that graph-based learning can recover meaningful spatial structure from expression and adjacency information [12,13]. Beyond unsupervised analyses, GNNs have also been applied to supervised prediction tasks in spatial cellular graphs. For example, SPACE-GM reports strong performance across multiple binary classification tasks, with ROC-AUC values above approximately 0.85, supporting the idea that neighborhood context carries predictive signal in tumor microenvironments [14,15]. This shows that there is a big area for improvement.

Data Set

We analyzed data from GSE174554, “A single cell atlas of human glioma under therapy,” a multi-assay glioblastoma dataset that includes single-nucleus RNA-seq (snRNA-seq) for 80 human IDH-wild-type GBM specimens (40 primary and 40 patient-matched recurrent) and a subset profiled with spatial transcriptomics. The study focuses on how standard-of-care therapy reshapes GBM cellular programs and the tumor microenvironment, providing both high-resolution transcriptomic profiles and spatially resolved measurements that support tumor-versus-normal labeling and downstream modeling.

For snRNA-seq and spatial data, we applied standard preprocessing (quality control, normalization, log transformation, dimensionality reduction) and constructed k-nearest-neighbor graphs for modeling. During harmonization between expression matrices and metadata, two samples were removed because they did not pass the combined QC and label-matching requirements needed for stable tumor–normal analysis.

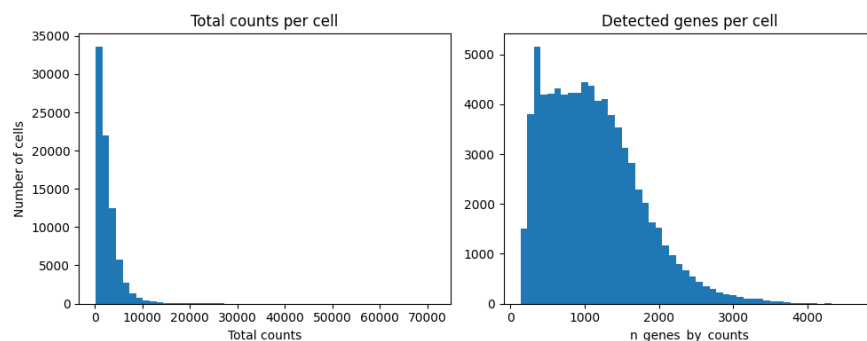


Fig 1: Distributions of total counts and detected genes per cell confirm that low-quality cells were removed while preserving high-quality profiles, justifying our thresholds and the 80k-cell working set.

Method

We analyzed tumor versus normal organization in glioblastoma using snRNA-seq and spatial transcriptomics. Data from each modality were preprocessed in Python, with cell–cell communication inference subsequently performed in R to enable comparison across multiple methods. These outputs were used to construct graph representations for GNN-based learning, which supported tumor–normal classification. CCC methods and benchmarking metrics were defined to ensure consistent and interpretable comparisons across algorithms and data modalities.

1. Data Preprocessing in Python

Processed expression matrices and associated metadata were loaded for each modality. Observation identifiers were matched to ensure consistent tumor and non-tumor labeling. Two samples were excluded due to failed label matching after filtering. Quality control was performed by computing standard metrics, including total counts and the number of detected genes per observation. Low-quality observations were removed based on these distributions to reduce noise from low-complexity or damaged profiles. Genes with low detection frequency were also filtered to reduce sparsity and improve downstream stability.

Counts were normalized to correct for library-size differences and log-transformed to stabilize variance. Highly variable genes were identified, and principal component analysis was applied to obtain a reduced representation of each observation. A k-nearest-neighbor graph was constructed in PCA space and used for downstream analyses. UMAP was computed for visualization, and Leiden clustering was performed on the same graph as a structural sanity check to verify that tumor and non-tumor labels were coherently distributed.

For spatial transcriptomics, spatial coordinates were retained for tissue-level visualization. Predicted tumor probabilities were mapped to spatial locations, and Gaussian smoothing was applied for visualization to highlight spatially coherent regions.

2. Data Preprocessing in R for CCC

We performed the CCC workflow in R because the Python environment did not allow stable execution of the CCC tools we intended to use. We encountered repeated dependency and environment conflicts, and several CCC packages we needed were either R-only or most reliable in R. To ensure we could run multiple CCC methods reproducibly and complete the benchmarking, we rebuilt the preprocessing and CCC-ready objects in R.

We created Seurat objects and computed standard QC metrics, including total counts, detected genes, and mitochondrial fraction. We applied the same filtering thresholds across samples, keeping observations with at least 200 detected genes, at most 6000 detected genes, and at most 15% mitochondrial reads. We also removed genes detected in fewer than 5 observations.

To control runtime and memory use, we downsampled very large samples and applied a global cap of 100,000 observations in the merged object used for downstream steps. We then normalized the data and selected 5,000 highly variable genes to reduce noise and stabilize later analyses.

We assigned broad cell-type labels using a reference-based annotation approach and required at least 50 observations per label. Labels below this threshold were merged into an “Other” category to avoid unstable groups. Finally, we joined tumor versus normal metadata to the Seurat objects so each retained nucleus or spot had a consistent tumor/normal label for downstream analysis.

3. CCC methods used

We inferred cell–cell communication under tumor versus normal grouping using three complementary approaches. Each method produces a ranked set of directed interactions from a sender group to a receiver group, annotated by ligand–receptor pairs and an associated score or probability. Using multiple methods was important because CCC inference can vary depending on the scoring model and filtering strategy, and our goal was to quantify which conclusions were stable versus method-dependent.

The first approach was a baseline ligand–receptor scoring method. For each ligand–receptor pair, we summarized expression within each group and computed an interaction score based on ligand expression in the sender group and receptor expression in the receiver group. This approach is

simple and transparent, and it tends to generate a dense interaction set, which makes it useful as a reference for network size and ranking comparisons.

The second approach was CellChat, a widely used CCC framework that combines curated ligand–receptor knowledge with a structured inference pipeline. CellChat performs overexpression and interaction filtering and estimates communication probabilities rather than relying only on a direct expression product. In practice, this often yields a smaller and more conservative set of interactions compared with baseline scoring, which helps test how robust the strongest signals are to stricter modeling assumptions.

The third approach was a SingleCellSignalR-inspired proxy method, reported as scsrproxy in our outputs. This method follows the same core idea of scoring interactions from ligand and receptor expression, but provides an independent implementation and ranking procedure. Including this third approach helps distinguish effects specific to one implementation from those consistent across baseline-style scoring methods.

4. CCC benchmarking

To compare CCC results across methods and across modalities, we standardized each CCC output into the same format, a directed edge list with sender group, receiver group, ligand, receptor, and an interaction score or probability. We then quantified CCC behavior using network size and agreement metrics. Network size was measured as the number of unique ligand–receptor pairs and the number of directed edges after aggregation. To measure agreement between CCC outputs, we computed Jaccard similarity between interaction sets:

$$J(A, B) = \frac{|A \cap B|}{|A \cup B|}$$

We calculated Jaccard overlap both for the full set of inferred ligand–receptor pairs and for the top-ranked edges (for example, the top 500), which focuses the comparison on high-confidence interactions. This benchmarking framework shows two key things. First, it reveals whether a method tends to produce dense versus conservative CCC networks through differences in pair and edge counts. Second, it quantifies how stable the inferred signaling landscape is across methods and modalities. Low overlap indicates strong method dependence, meaning biological conclusions should be reported as sensitive to the inference approach rather than treated as a single definitive CCC network.

5. Graph construction and GNN training

We modeled tumor vs. normal prediction as a node classification task on graphs built separately for snRNA-seq and spatial transcriptomics. In both modalities, each observation was treated as a node. For snRNA-seq, nodes represent individual nuclei. For spatial transcriptomics, nodes represent spatial spots. Node features were derived from the processed expression representation used for modeling after normalization and dimensionality reduction.

We defined edges using a k-nearest-neighbor graph in the learned latent space. Two nodes are connected if they are among each other's nearest neighbors in this space, which captures local similarity structure. This edge definition is useful for both modalities. In snRNA-seq, it links cells with similar transcriptional states, which helps smooth noisy expression and supports classification in heterogeneous tumor populations. In spatial transcriptomics, it links spots with similar expression profiles and often aligns with local tissue structure, which encourages the model to learn region-level patterns rather than isolated spot-level noise.

For the classifier, we used a GraphSAGE-style GNN. The model consists of stacked neighborhood-aggregation layers that update each node's embedding by combining the node's features with information from its neighbors. The final node embedding is passed to a fully connected classification head that outputs the probability of the tumor class. A separate architecture figure in the report summarizes the input features, GraphSAGE layers, and classifier head.

We trained the model using a binary cross-entropy loss for tumor versus normal labels. We split the data into 60% for training, 20% for validation, and 20% for testing. Training used the training set, model selection used the validation set, and final results were reported on the held-out test set. We evaluated performance using Accuracy, Precision, Recall, F1 Score, and ROC-AUC. For spatial transcriptomics, we also mapped predicted tumor probabilities back to tissue coordinates. We visualized them as a probability scatter plot and a Gaussian-smoothed probability surface to assess spatial coherence and interpretability.

Results

1. Data quality and tumor label visualization

We first checked data quality and confirmed that tumor versus normal labels behaved as expected after preprocessing. The QC distributions showed typical variability in library size and detected genes, which justified filtering low-quality observations before downstream analysis. After normalization and dimensionality reduction, the UMAP embedding showed clear structure, and tumor versus normal labels were not confined to a single isolated region. This suggested that tumor classification would not be a trivial clustering artifact and that neighborhood-based learning could be informative.



Figure 2. A. UMAP of the processed snRNA-seq (80k cells, 5k genes) colored by Leiden clusters shows clean, well-separated groups, indicating successful QC and capturing the expected heterogeneity of GBM. **B.** The same UMAP colored by tumor vs non-tumor labels shows tumor cells forming coherent regions and non-tumor cells mapping to distinct neighborhoods, supporting a credible sender/receiver setup for communication analysis.

2. CCC outputs under tumor versus normal grouping

We inferred tumor–normal CCC using three approaches. Baseline ligand–receptor scoring, CellChat, and a SingleCellSignalR-style proxy method scsrproxy. Because CCC tools were not stable in our Python environment, we ran CCC in R to ensure we could execute multiple methods and export comparable outputs. Each method produced a ranked list of directed tumors to normal and normal to tumor interactions annotated by ligand–receptor pairs.

A major result is that the CCC network size differed strongly by method. In snRNA-seq under tumor–normal grouping, baseline scoring produced a dense interaction set with 671 unique ligand–receptor pairs and 2,684 directed edges, while CellChat produced a much smaller set with 13 unique pairs and 14 directed edges. This gap indicates that stricter modeling and filtering can heavily reduce the inferred tumor–normal communication network, which changes which interactions are available for interpretation.

modality	grouping	method	n_pairs	n_edges
snRNA	celltype	baseline	671	13420
snRNA	celltype	cellchat	132	1574
snRNA	tumor	baseline	671	2684
snRNA	tumor	cellchat	13	14
spatial	tumor	baseline	73	657
spatial	tumor	cellchat	24	59

Table 1. CCC network size across modalities and methods. We report the number of unique ligand–receptor pairs (n_pairs) and the number of directed edges (n_edges) for snRNA-seq and spatial transcriptomics under celltype and tumor–normal grouping

3. CCC benchmarking and agreement across methods

To quantify how similar or different CCC outputs were, we benchmarked the overlap of inferred interaction sets. We reported Jaccard similarity on the set of ligand–receptor pairs and on the top-ranked edges, where the top-edge comparison focuses on high-confidence predictions. The agreement results showed two clear patterns. First, baseline and scsrproxy were highly consistent

with each other on top-ranked edges, which is expected because both are scoring-based approaches. Second, CellChat had very low overlap with baseline-style outputs, meaning the “top interactions” depended strongly on which CCC method was used. This is the main reason benchmarking mattered in our pipeline. It prevents us from presenting a single CCC network as the only answer.

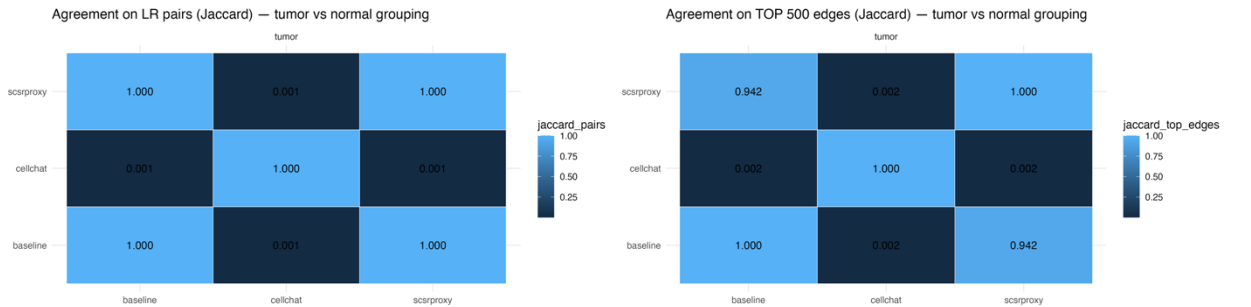


Figure 3A (LR pair agreement). Jaccard similarity heatmap comparing CCC methods on inferred ligand–receptor pair sets under tumor–normal grouping. **3B** (Top-edge agreement). Jaccard similarity heatmap comparing CCC methods on the top-ranked edges under tumor–normal grouping.

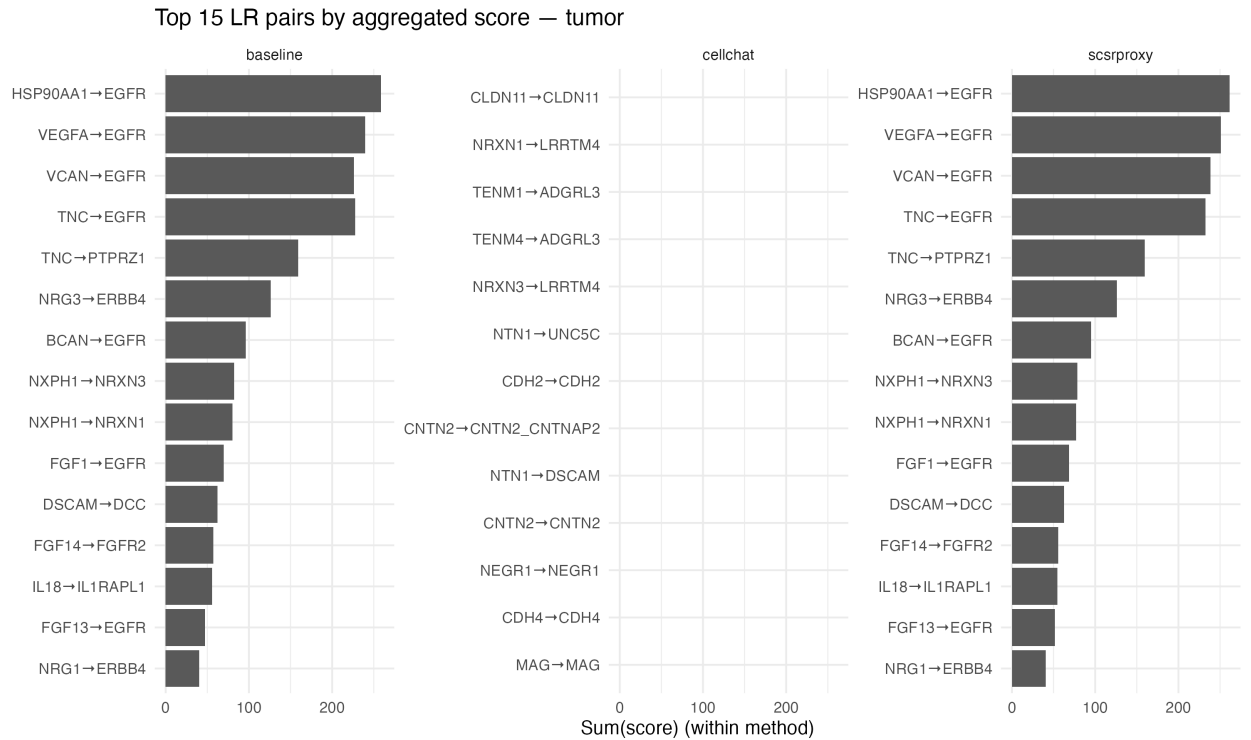


Figure 4. (Top LR pairs). Bar plot of the top-ranked ligand–receptor pairs under tumor–normal grouping, shown by method.

grouping	method	consensus_pairs	consensus_top_edges	n_edges	n_pairs	consensus_overall
celltype	baseline	0.510814249	0.473703601	13420	671	0.492258925

celltype	scsrproxy	0.510814249	0.473190863	13420	671	0.492002556
celltype	cellchat	0.021628499	0.012658487	1574	132	0.017143493
tumor	baseline	0.500732064	0.471848445	2684	671	0.486290255
tumor	scsrproxy	0.500732064	0.471848445	2684	671	0.486290255
tumor	cellchat	0.001464129	0.001949318	14	13	0.001706723

Table 2. CCC benchmarking agreement across methods. consensus pairs and consensus_top_edges summarize overlap with the consensus interaction set using Jaccard similarity for ligand–receptor pairs and top-ranked edges. consensus_overall reports an overall agreement summary per method within each grouping.

4. Graph neural network tumor classification

We trained GraphSAGE-style GNN classifiers on graphs built from snRNA-seq and spatial transcriptomics. Nodes represented nuclei or spots, and edges came from k-nearest-neighbor connectivity in latent space. We used a 60/20/20 train/validation/test split and optimized binary cross-entropy loss. Final evaluation used Accuracy, Precision, Recall, F1, and ROC-AUC on the held-out test set.

The snRNA-seq model achieved strong performance with an Accuracy of 0.880 and an ROC-AUC of 0.949. The spatial model achieved higher threshold-based performance with an Accuracy of 0.947 and an F1 of 0.889, and it showed very conservative tumor calling with a Precision of 1.0. At the same time, the spatial ROC-AUC was lower (0.871), which is consistent with a limited spatial sample size and higher spot-level noise. Overall, these results support graph-based tumor versus normal prediction in both modalities. They are also consistent with prior work showing that graph-based models can achieve strong ROC-AUC in spatial prediction settings, often above 0.85, although direct comparisons are not strict because tasks and datasets differ.

Modality	Accuracy	Precision	Recall	F1 Score	ROC-AUC
snRNA-seq	0.8803	0.8702	0.7859	0.8259	0.9489
Spatial	0.9474	1.0000	0.8000	0.8889	0.8714

Table 3. Tumor versus normal node classification performance for GNN models trained on snRNA-seq and spatial transcriptomics graphs. Metrics are reported on the held-out test set.

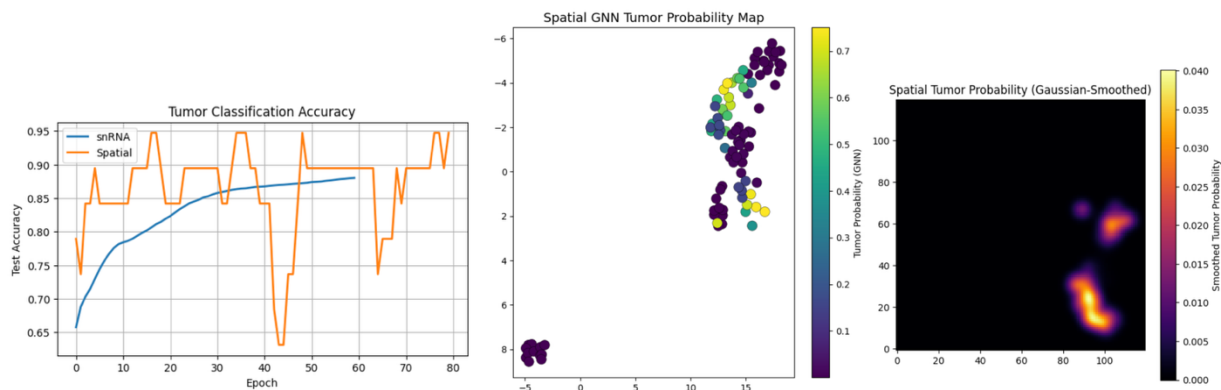


Figure 5A (Training curve). Test accuracy across epochs for snRNA-seq and spatial models. 5B (Spatial probability map). Spot-level tumor probability mapped to tissue coordinates. 5C

(Smoothed spatial map). Gaussian-smoothed tumor probability map to highlight coherent regions.

Limitations

The main limitation in the spatial setting is sample scale. Spatial transcriptomics in this dataset is available for a smaller subset than snRNA-seq, and spots can mix signals from multiple cell types. This reduces the amount of clean supervised signal available to the spatial model and can reduce ROC-AUC even when accuracy looks strong at a fixed threshold. A second limitation is CCC tool coverage in Python. We originally aimed to run more CCC algorithms in one pipeline, but environment conflicts and R-only package availability forced us to move CCC inference to R. This choice improved reproducibility and allowed multi-method benchmarking, but it reduced the number of Python-based CCC methods we could include in a single unified runtime.

Conclusion

In this study, we analyzed glioblastoma tumor versus normal structure using matched snRNA-seq and spatial transcriptomics from GSE174554. We combined two complementary analyses. First, we trained graph neural networks on modality-specific graphs to classify tumor versus normal. Second, we inferred tumor–normal cell-cell communication using multiple CCC approaches and benchmarked how strongly the inferred interaction networks depended on the method.

Our modeling results show that graph-based learning can separate tumors from normal tissue with strong performance in both modalities. The snRNA-seq model achieved high discriminative ability, while the spatial model achieved higher accuracy and a very conservative tumor calling behavior, which was also visually supported by coherent high-probability regions on tissue coordinates. These findings support the value of representing single-cell and spatial transcriptomic data as graphs, where neighborhood structure helps stabilize predictions in noisy and heterogeneous data.

Our CCC benchmarking results demonstrate that tumor–normal CCC conclusions are method-dependent. Baseline-style scoring approaches produced dense interaction sets and showed strong agreement with each other, while CellChat produced a much smaller set of inferred interactions and low overlap with baseline-style outputs. This highlights an important practical point: CCC outputs should not be treated as a single definitive network without benchmarking, especially when group definitions and filtering decisions can strongly change which edges are retained.

Overall, this work provides a reproducible tumor–normal pipeline that links graph-based prediction with CCC benchmarking in GBM. Future work should expand the CCC tool coverage using containerized environments, add confusion matrices and repeated-seed evaluation for the GNN models, and incorporate spatially defined adjacency directly as an alternative to latent-space neighbor graphs.

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Extra graphs:

